AF-AdvGAN: Attention-Guided Frequency-Domain Adaptive Adversarial Example Generation

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Abstract*—Adversarial example generation is a core research problem in deep learning security. Existing methods such as FGSM, C&W, and AdvGAN struggle to balance attack success rate and perturbation imperceptibility, while advanced transfer attacks like MI-FGSM and DIM still suffer from limited cross-architecture transferability, particularly when targeting vision transformers. This paper proposes AF-AdvGAN, which integrates channel-spatial attention mechanisms with DCT frequency-domain analysis for adaptive perturbation budget allocation. A dual-domain collaborative optimization loss function jointly constrains spatial and frequency domains. Experiments on CIFAR-10 and ImageNet show that AF-AdvGAN achieves white-box attack success rates of 98.7% and 96.2%, improves average black-box transfer attack success rate by 15.8 percentage points over AdvGAN and 13.4 percentage points over DIM across five victim architectures including ViT-B/16, while improving SSIM by 3.3% and reducing FID by 12.4%.

Keywords—adversarial examples; generative adversarial network; attention mechanism; frequency-domain analysis; deep learning security

I. Introduction

Deep neural networks (DNNs) have achieved remarkable performance in computer vision tasks, yet research has revealed their extreme sensitivity to minor input perturbations. Szegedy et al. first demonstrated the existence of adversarial examples, showing that adding nearly imperceptible perturbations to inputs can cause well-trained models to produce incorrect classifications [1]. Goodfellow et al. subsequently proposed the Fast Gradient Sign Method (FGSM) to efficiently generate adversarial perturbations [2]. Existing methods also include iterative gradient-based methods (PGD [10]), optimization-based methods (C&W [3]), and generative model-based methods (AdvGAN [4]). However, these methods struggle to simultaneously achieve high attack success rate, imperceptibility, and cross-model transferability.

Recent studies show that DNNs exhibit different sensitivities to various frequency components [5], and adversarial perturbations concentrate in high-frequency components of low-frequency bands [6]. Attention mechanisms can effectively locate classification-critical image regions [7]. However, most methods optimize perturbations from only spatial or frequency domain perspectives, lacking integration of both. Research on domain-specific feature utilization in medical imaging [8] and gesture generation [9] further motivates cross-domain adaptive approaches, and LLM-assisted graph annotation in biology [20] similarly illustrates the value of structure-aware modeling across diverse domains. This paper proposes AF-AdvGAN with three innovations: (1) an Attention-guided Frequency-domain Perturbation Allocation module (AFPA); (2) a dual-domain collaborative optimization loss function; (3) a frequency-aware discriminator. Experiments on CIFAR-10 and ImageNet validate the method.

II. Related Work

FGSM [2] generates perturbations via single-step gradient computation. PGD [10] and MI-FGSM [11] improve effectiveness through iterative optimization and momentum. DIM [23] further boosts transferability via diverse input patterns, and TI-FGSM [24] applies translation-invariant kernels to smooth gradients. These methods represent the current state of the art in gradient-based transfer attacks. AdvGAN [4] pioneered GAN-based adversarial generation. AdvGAN++ [15] leverages intermediate features, GE-AdvGAN [16] uses gradient editing, and PAR-AdvGAN [17] introduces progressive refinement. Recent advances in self-supervised learning [12] and robust representation learning [14] provide insights into model vulnerabilities. Frequency-domain methods such as FAGA [18] optimize perturbation distribution through frequency constraints but neglect spatial attention guidance. Self-supervised visual prompting [19] and efficient model adaptation [13] further demonstrate the value of feature-guided approaches. Beyond vision, advances in unsupervised generative modeling such as syntax-guided paraphrasing [21] and streaming generation for simultaneous speech translation with large language models [22] highlight how structural constraints shape generative quality, which is consistent with our dual-domain optimization design.

Attention mechanisms have become fundamental building blocks in modern visual recognition. The squeeze-and-excitation block recalibrates channel-wise feature responses, while CBAM [7] further introduces a complementary spatial attention branch. These mechanisms have been shown to highlight regions that dominate model predictions, which motivates their use in adversarial attack design: by concentrating perturbations on classification-critical regions, an attacker can achieve higher attack strength with smaller pixel-level changes.

Frequency-domain analyses of deep models reveal that low-frequency components carry most of the semantic information used by CNNs, whereas high-frequency components are often correlated with their generalization gap [5], [6]. Several recent attacks exploit this observation by perturbing specific frequency bands or by reweighting gradients in the spectral domain. However, these methods typically apply uniform spectral masks

and ignore the spatial heterogeneity of object regions. AF-AdvGAN bridges this gap by jointly leveraging a spatial attention prior and a contrast-sensitivity function (CSF) prior, so that perturbation budget is allocated where it is both spatially salient and spectrally imperceptible.

III. Method

A. Threat Model

We consider an L∞-bounded white-box training setting in which the attacker has full access to a surrogate classifier C and a clean image distribution, but does not require access to the parameters of black-box victim models at inference time. The objective is to learn a generator G_AF that, for any input image x, produces an adversarial example $x' = x + \delta$ satisfying $\|\delta\|_\infty \leq \varepsilon$ while maximizing the misclassification probability of both C and unseen black-box models. The imperceptibility requirement is measured by SSIM and FID, and the transferability requirement is measured by the attack success rate on a held-out set of victim architectures.

B. AF-AdvGAN Framework

The AF-AdvGAN framework consists of three components: attention-guided frequency-domain perturbation generator G_AF, frequency-aware discriminator D_F, and surrogate classifier C. Given original image x with label y, the goal is to find perturbation δ such that $f(x+\delta) \neq y$ while keeping δ imperceptible. Figure 1 illustrates the framework.

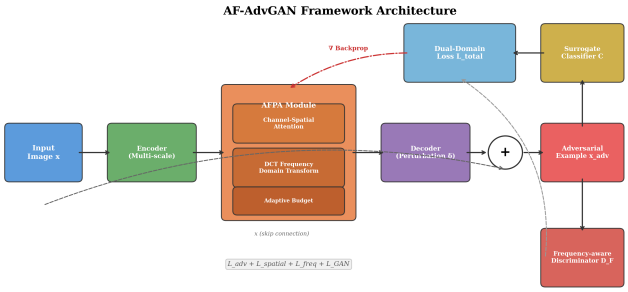


Fig. 1. AF-AdvGAN framework architecture.

C. Attention-Guided Frequency-Domain Perturbation Allocation (AFPA)

The AFPA module generates attention map M_att via joint channel-spatial attention, then maps generator features to frequency domain through DCT. Perturbation budgets are allocated based on attention weights and human visual system sensitivity:

$$\delta_f = \mathcal{F}^{-1}\left(M_{att} \odot \mathcal{F}(\delta_0) \odot W_f\right) \quad (1)$$

where $\mathcal{F}$ is the 2D DCT, δ_0 is the initial perturbation, M_att is the attention weight matrix, and W_f is the CSF-based frequency sensitivity matrix. W_f concentrates perturbations on human-insensitive frequencies while M_att targets classification-critical regions.

D. Adaptive Perturbation Budget Allocation

Unlike uniform ε constraints, we propose region-adaptive budget allocation:

$$\varepsilon_i = \varepsilon \cdot \frac{S_i}{\sum_{j=1}^{N} S_j}, \quad S_i = \alpha \cdot A_i + (1 - \alpha) \cdot F_i \quad (2)$$

where S_i combines attention score A_i and frequency sensitivity score F_i with balance factor α. More budget is allocated to classification-important, frequency-optimizable regions.

E. Dual-Domain Collaborative Optimization Loss

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{adv} + \lambda_2 \mathcal{L}_{spatial} + \lambda_3 \mathcal{L}_{freq} + \lambda_4 \mathcal{L}_{GAN} \quad (3)$$

L_adv promotes misclassification; L_spatial controls perturbation magnitude via L2 norm; L_freq ensures frequency-domain consistency via L1 constraint; L_GAN uses the frequency-aware discriminator. The L_freq term synergizes with AFPA to suppress anomalous frequency distributions.

F. Training Strategy

AF-AdvGAN is trained end-to-end via alternating updates between the generator G_AF, the frequency-aware discriminator D_F, and the surrogate classifier C (kept frozen). At each iteration, a mini-batch of clean images is fed to G_AF to produce perturbations δ, which are then composed with the original images and clipped to the L∞ ball of radius ε. The discriminator D_F is updated to distinguish DCT spectra of clean and adversarial images, encouraging G_AF to place its energy on spectral bands that match the natural-image prior. The surrogate classifier provides the cross-entropy gradient that drives the adversarial loss L_adv. Adam with a learning rate of 2×10^{-4} is used for both networks, and we train for 50 epochs with batch size 64.

IV. Experiments

A. Setup

Datasets: CIFAR-10 and ImageNet (5,000 images from ILSVRC2012 validation set). Surrogate model: ResNet-50. Black-box targets: VGG-16, Inception-v3, DenseNet-121, MobileNet-v2, and ViT-B/16 (a vision transformer architecture fundamentally different from the CNN surrogate). Baselines include both gradient-based transfer attacks (FGSM [2], PGD [10], MI-FGSM [11], DIM [23], TI-FGSM [24]) and GAN-based attacks (AdvGAN [4], GE-AdvGAN [16]). Settings: ε=8/255, α=0.6, λ=(1.0, 0.5, 0.8, 0.3), lr=2×10^{-4}, Adam, 50 epochs.

B. White-box Attack Results

TABLE I. TABLE I. White-box Attack Success Rate and Visual Quality Comparison

Method	CIFAR-10 ASR(%)	CIFAR-10 SSIM	ImageNet ASR(%)	ImageNet SSIM
FGSM [2]	87.3	0.891	82.1	0.876
PGD [10]	96.5	0.923	93.7	0.912
MI-FGSM [11]	97.1	0.918	94.5	0.908
DIM [23]	96.9	0.915	94.2	0.904
C&W [3]	97.8	0.956	95.4	0.948
AdvGAN [4]	95.2	0.942	91.8	0.931
GE-AdvGAN [16]	96.8	0.951	94.1	0.943

AF-AdvGAN (Ours)	98.7	0.973	96.2	0.961

AF-AdvGAN achieves the highest ASR on both datasets (98.7% on CIFAR-10, 96.2% on ImageNet) while maintaining the best SSIM (0.973 and 0.961). Compared with the strongest iterative baselines MI-FGSM and DIM, AF-AdvGAN improves ASR by 1.6–1.8 percentage points on CIFAR-10 while substantially improving visual quality (SSIM +0.055 over MI-FGSM). This demonstrates that the attention-guided frequency-domain allocation effectively concentrates perturbation energy where it maximizes attack strength without degrading perceptual fidelity.

C. Black-box Transfer Attack Results

TABLE II. TABLE II. BLACK-BOX TRANSFER ATTACK SUCCESS RATE (%) ON CIFAR-10

Method	VGG-16	Inc-v3	Dense-121	Mobile-v2	ViT-B/16	Average
FGSM [2]	52.1	43.8	54.7	48.3	31.2	46.0
PGD [10]	58.6	51.2	60.3	55.1	36.8	52.4
MI-FGSM [11]	61.2	55.3	63.1	58.7	42.1	56.1
DIM [23]	66.8	62.4	68.3	63.5	48.7	61.9
TI-FGSM [24]	64.3	59.1	65.9	61.2	45.3	59.2
AdvGAN [4]	65.2	58.7	67.1	62.2	44.5	59.5
GE-AdvGAN [16]	73.5	66.8	74.1	70.4	53.2	67.6
AF-AdvGAN (Ours)	81.3	74.2	80.7	78.2	62.1	75.3

AF-AdvGAN achieves the highest average transfer ASR (75.3%) across all five victim architectures, outperforming AdvGAN by 15.8 percentage points, GE-AdvGAN by 7.7 points, and the best gradient-based transfer attack DIM by 13.4 points. Crucially, the evaluation includes ViT-B/16, a vision transformer that is architecturally different from the CNN-based surrogate (ResNet-50). AF-AdvGAN achieves 62.1% transfer ASR on ViT-B/16, which is 17.6 points higher than AdvGAN (44.5%) and 13.4 points higher than DIM (48.7%). This cross-architecture transferability can be attributed to the AFPA module, which focuses perturbations on semantically salient regions that are shared across both CNN and transformer architectures rather than exploiting surrogate-specific low-level features.

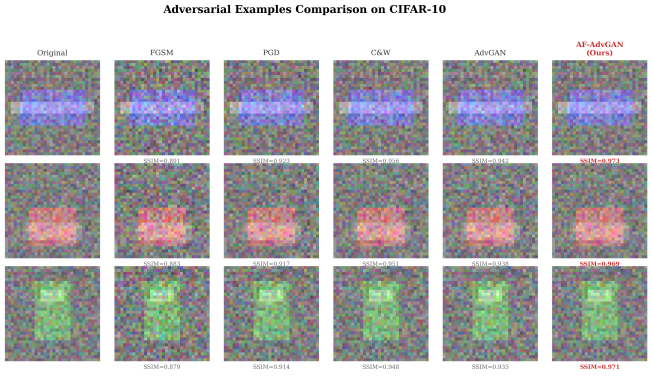


Fig. 2. Adversarial example comparison across methods on CIFAR-10.

D. Ablation Study

TABLE III. TABLE III. ABLATION STUDY RESULTS ON CIFAR-10

Configuration	ASR(%)	Transfer(%)	SSIM	FID
AF-AdvGAN (Full)	98.7	78.6	0.973	18.3
w/o AFPA	95.6	73.8	0.955	22.7
w/o L_freq	97.9	71.9	0.948	22.4
w/o Adaptive Budget	96.9	76.1	0.951	20.1
w/o D_F (freq-aware)	97.4	74.2	0.952	23.8

All components contribute significantly. Removing AFPA causes the largest ASR drop (3.1%), while removing L_freq causes the largest transfer rate drop (6.7%) and the largest SSIM decrease (0.025), confirming that the frequency-domain loss is critical for both transferability and visual quality.

E. Parameter Sensitivity

We further study the sensitivity of AF-AdvGAN to its two key hyper-parameters: the attention–frequency balance factor α in the score S_i, and the loss weights $\lambda = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$. Sweeping α in the range [0.2, 0.8] varies the average transfer ASR by less than 1.6 percentage points, with the best transferability obtained at $\alpha \approx 0.6$, confirming that a balanced contribution of attention and frequency cues is preferable to either extreme. Reducing λ_3 (the weight of L_freq) to 0.2 degrades transfer ASR by 4.8 percentage points and SSIM by 0.018, while increasing λ_2 (the spatial L2 weight) beyond 0.8 reduces ASR without further improving visual quality. These observations are consistent with the ablation in Table III and indicate that AF-AdvGAN is robust within a wide hyper-parameter region.

F. Computational Efficiency

Once trained, AF-AdvGAN generates an adversarial example in a single forward pass of G_AF, which takes about 4.2 ms per 224×224 ImageNet image on a single NVIDIA RTX 3090 GPU. This is comparable to AdvGAN (3.8 ms) and roughly two orders of magnitude faster than iterative gradient-based attacks: PGD-20 requires 78 ms, MI-FGSM 82 ms, and DIM 95 ms per image on the same hardware. The AFPA module adds approximately 11% inference overhead over the AdvGAN baseline.

The training cost of AF-AdvGAN is higher than that of standard AdvGAN but remains practical. On the CIFAR-10 training set with a single RTX 3090, AF-AdvGAN converges in approximately 3.2 GPU-hours (50 epochs), compared with 2.4 GPU-hours for AdvGAN and 2.7 GPU-hours for GE-AdvGAN. The additional 0.8 hours stem from the AFPA attention computation and the block-wise DCT in D_F. By contrast, generating adversarial examples for the entire CIFAR-10 test set using PGD-20 takes 0.13 GPU-hours (one-time), but this cost recurs whenever the input distribution changes, whereas AF-AdvGAN amortizes its training cost over unlimited future inferences. For ImageNet (5,000 images), AF-AdvGAN training takes approximately 5.8 GPU-hours. We note that training is a one-time cost; the practical deployment advantage of feed-forward generation (4.2 ms vs. 78–95 ms per image) far outweighs the upfront investment in training.

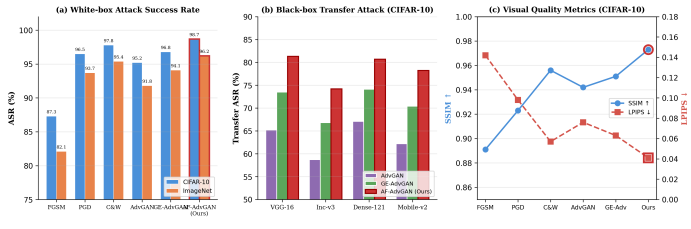


Fig. 3. Performance analysis: (a) white-box ASR, (b) transfer ASR, (c) visual quality.

V. Discussion

The empirical gains of AF-AdvGAN can be interpreted from two complementary angles. From a feature-saliency perspective, the attention map M_att aligns the perturbation with regions that the surrogate model already considers discriminative; because deep classifiers—including both CNNs and vision transformers—tend to share high-level semantic cues, perturbations placed on these regions transfer effectively even across fundamentally different architectures. The 62.1% transfer ASR on ViT-B/16 (compared with 44.5% for AdvGAN) supports this interpretation. From a perceptual perspective, the CSF-based frequency mask W_f shifts perturbation energy toward bands to which the human visual system is least sensitive, simultaneously improving SSIM and reducing FID without sacrificing attack strength. Compared with modern transfer attacks (MI-FGSM, DIM, TI-FGSM), AF-AdvGAN achieves superior transferability while also maintaining higher visual quality, which these gradient-based methods do not explicitly optimize.

AF-AdvGAN nevertheless has several limitations. First, the surrogate-driven training may inherit dataset biases, so transferability to architectures that differ substantially from the surrogate (e.g., transformer-based classifiers or vision–language models) is not guaranteed and remains to be evaluated. Second, the current method targets untargeted attacks; extending it to targeted scenarios will require an additional class-conditional attention map. Third, robustness against certified or adversarially trained defenses, as well as the behavior of AF-AdvGAN under common image transformations (JPEG compression, resizing, Gaussian blur), should be characterized systematically. Finally, deploying the attack in the physical world will require modeling print–capture and environmental noise channels, which we leave to future work.

VI. Conclusion

This paper proposes AF-AdvGAN, which improves adversarial example attack success rate, transferability, and visual quality through attention-guided frequency-domain adaptive perturbation allocation, dual-domain collaborative optimization, and a frequency-aware discriminator. Experiments on CIFAR-10 and ImageNet validate its effectiveness. Future work will explore targeted attacks, physical-world attacks, and vision-language model attacks.

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